

The Quality of Speculation: Common Ground for Speculative Design in Human-Computer Interaction?

Ronda Ringfort-Felner
Ubiquitous Design / Experience &
Interaction
University of Siegen
Siegen, Germany
ronda.ringfort@uni-siegen.de

Judith Dörrenbächer
Ubiquitous Design / Experience &
Interaction
University of Siegen
Siegen, Germany
judith.doerrenbaecher@uni-siegen.de

Chris Elsdén
Institute for Design Informatics
University of Edinburgh
Edinburgh, United Kingdom
chris.elsden@ed.ac.uk

James H. Auger
CRD
École normale supérieure Paris-Saclay
Paris, France
james.auger@ens-paris-saclay.fr

James Pierce
School of Art + Art History + Design
University of Washington Seattle
Seattle, Washington, USA
jppierce@uw.edu

Richmond Y. Wong
Digital Media
Georgia Institute of Technology
Atlanta, Georgia, USA
rwong34@gatech.edu

Marc Hassenzahl
Ubiquitous Design / Experience &
Interaction
University of Siegen
Siegen, Germany
marc.hassenzahl@uni-siegen.de

Abstract

Speculative design is increasingly being used in Human-Computer Interaction (HCI) to explore the not-yet. Nonetheless, it remains methodologically vague and is often contested. Especially, what counts as good speculation, and how its quality can be conceptualized, is an ongoing discussion. This workshop brings together researchers, designers, and practitioners to collectively explore questions of quality in speculative design. Building on a recently published descriptive taxonomy of quality, the workshop offers this framework as an initial common ground for shared reflection and critique. Participants will analyze speculative designs and engage in critical discussion to examine how quality is constructed and negotiated across practices and contexts. By challenging the taxonomy's applicability and limits, the workshop will explore diverse perspectives on what constitutes quality. Insights from the workshop will inform refinements to the taxonomy, deepen the conceptualization of quality in speculative design, and strengthen its legitimacy as a research practice.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models.

Keywords

Speculative Design, Quality, Evaluation, Method

ACM Reference Format:

Ronda Ringfort-Felner, Judith Dörrenbächer, Chris Elsdén, James H. Auger, James Pierce, Richmond Y. Wong, and Marc Hassenzahl. 2026. The Quality of Speculation: Common Ground for Speculative Design in Human-Computer Interaction?. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3778760>

1 Background and Motivation

Speculative design has become a key approach in Human-Computer Interaction (HCI) for exploring ethical, political, and societal implications of technologies that do not yet exist [35]. Research and design practices such as design fiction [6, 8, 32], critical design [2, 13], speculative enactments [11], and material speculation [33] enable researchers to question current technological trajectories and imagine alternative presents and futures. They translate abstract imaginaries into tangible forms such as narratives, fictional research papers, prototypes, videos, or enactments. Unlike approaches concerned with market feasibility or product implementation, these practices seek to provoke critical reflection and debate about possible futures [1, 10, 24, 26].

What began in art and design studios has steadily moved into academic discourse. In HCI, speculative design now functions as “research *through* design” [14], where it is primarily used to generate knowledge about the future [18]. In this context, it is required to meet the demands of a scientific approach for argumentative grounding and evaluative consistency. From publications and funding to teaching and collaboration, authors, reviewers, and educators must increasingly articulate and evaluate what speculative design contributes.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '26, Barcelona, Spain

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3778760>

To strengthen its legitimacy as a mode of research, scholars have proposed definitions (e.g., [13, 19, 27]), classifications (e.g., [7, 16, 17, 26]), taxonomies (e.g., [12, 15, 22]), specific methods (e.g., [21, 23, 25, 30, 34]) and initial evaluative considerations (e.g., [3, 5, 9, 18]). While these efforts have clarified speculative design's position in HCI, they leave key questions open. Chief among these is the question of *quality*: What counts as a *good* speculation, and how can its quality be conceptualized?

So far, discussions of quality in speculative design remain largely implicit, shaped more by individual interpretations than by shared understanding. This has significant consequences: Reviewers fall back on familiar HCI methods such as prototypes or user studies to evaluate speculations (e.g., [5, 20]), even though the implied quality aspects do not necessarily apply to speculative design [4]. Researchers create compelling speculations that fail to spark the intended depth of discussion (e.g., [4, 29]). Companies claim to use speculative design but often treat it as a buzzword, stripped of its critical edge [31]. The result is divergent notions of quality, inconsistent assessments in peer review, challenges in conducting effective speculative research, and persistent doubts about speculative design's legitimacy as a research practice.

1.1 Quality in Speculative Design

In response to these challenges, a recent scoping review examined 63 theoretical publications on speculative design, design fiction, and critical design [28]. It collated the widely scattered qualities scholars associate with successful speculation in HCI. The review identified nine recurring qualities, grouped in three categories: *Speculative qualities* (fictional, critical, socio-political) capture the core characteristics of speculative design. *Discursive qualities* (experienceable, thought-provoking) facilitate reflection and debate. *Process qualities* (grounded, participative, reflected, playful) encourage an inclusive, responsible, scientifically based and creative approach to speculative design. Together, these qualities form a *descriptive taxonomy* that serves as a starting point for creating and evaluating speculative designs. By making qualities, outcomes, and impacts explicit, the taxonomy invites situated justification: researchers and designers are encouraged to select and prioritize those most relevant to a speculation's purpose. Quality is thus positioned as something to be interpreted in relation to the knowledge a speculation aims to create [5].

This taxonomy marks an initial step toward a shared vocabulary but leaves key questions open: Do its criteria resonate with the understandings of those who practice and apply speculative design beyond the reviewed corpus? Can the taxonomy serve as a common ground for evaluation and justification across the broader HCI community? The taxonomy therefore opens rather than closes the debate: it provides a provisional common ground that calls for critical discussion and negotiation about what quality in speculative design means and how it can be made explicit.

1.2 Workshop Themes and Goals

The workshop provides a forum to critically examine and advance understanding of quality in speculative design. It is structured around three interconnected themes:

- **Understanding Quality:** Investigating how notions of quality in speculative design are constructed and negotiated across practices and contexts.
- **Testing the Taxonomy:** Assessing where the taxonomy provides guidance, where it falls short, and how it might be adapted.
- **Building Common Ground:** Exploring whether—and how—common ground for quality in speculative design can be achieved.

Together, these themes frame the workshop as a space for discussion and productive disagreement, with the broader goal of strengthening speculative design as a legitimate research practice. Specifically, it aims to:

- make implicit notions of quality explicit for collective reflection,
- strengthen a community to negotiate and advance quality in speculative design,
- refine and situate understandings of quality to advance its conceptualization in speculative design (e.g., through recommendations or refinements to the taxonomy) and how it relates to other approaches in HCI research.

Unlike other workshops on speculative design that focus on creating and investigating specific futures or artefacts, this workshop turns the focus to *methodology* and *quality*. Its concern is not to generate speculations, but to co-create the vocabulary needed to justify their value.

2 Workshop Structure and Activities

The workshop is planned as a **long workshop** consisting of two sessions with a break. We aim for **12–15 participants**, balancing diversity of perspectives with a size that supports intensive group work and collective synthesis.

2.1 Before the Workshop

Participants will submit a **speculative design** (e.g., fictional research paper, podcast, prototype, magazine, scenario description, or a documentation of a speculative enactment) together with a **short personal reflection** on what makes it strong or weak as a speculation in relation to its intended goals.

2.2 Session 1: Framing and Analysis (90 min)

- **Welcome & Introduction (5 min):** Organizers open the workshop with a brief welcome and an overview of its goals and agenda.
- **Lightning Participant Presentations (45 min):** Each participant briefly introduces themselves and presents their submitted speculation and accompanying reflection. Written materials and any physical or digital artifacts will be displayed around the room to enable later interaction.
- **Taxonomy Introduction (15 min):** Organizers provide a concise introduction to the taxonomy and its criteria, followed by a Q&A to ensure a shared understanding among participants.

- **Group Analysis–Part 1 (25 min):** In groups of 2–3, participants apply the taxonomy to one or two submitted speculations. The aim is to examine how the taxonomy shapes evaluation and whether its qualities align with or diverge from the author’s own criteria. Each group will create a short visual summary of their discussion (e.g., key points on a A2 sheet or sticky notes attached to the artifacts).

2.3 Session 2: Synthesis and Mapping (90 min)

- **Group Analysis–Part 2 (15 min):** Groups revisit and refine their analyses, optionally extending them to another speculation, and finalize their visual summaries.
- **Gallery Walk (20 min):** Participants circulate to read and annotate each group’s summaries, identifying connections, tensions, and recurrent issues across their evaluations, the authors’ reflections, and the speculations. Annotations are added directly to the visual summaries, either as written comments or as sticky notes.
- **Collective Mapping & Discussion (50 min):** Building on these annotations, participants will collaboratively cluster their observations. This process will lead into a plenary discussion on how quality is understood in speculative design and whether—and how—common ground for HCI can be achieved.
- **Closing (5 min):** Organizers summarize key insights and outline potential next steps.

2.4 Accessibility and Documentation

We strive to make the workshop accessible and inclusive. Authors of accepted submissions will be asked to meet accessibility standards (e.g., screen-reader-friendly PDFs, captions or transcripts for audio/video). With consent, submissions will be shared with participants in advance, alongside the descriptive taxonomy of nine quality criteria. This will allow participants to familiarize themselves with the materials, provide early feedback, and prepare for exchange during the session. Organizers will also consult participants beforehand to identify specific access needs and arrange accommodations (e.g., live captioning). During the workshop, we will also provide clear multimodal instructions (oral, written, visual) to support different communication preferences in group work.

The workshop will be documented through photographs (with consent), written notes, and digitized outputs such as refined quality criteria, exemplary cases, and adaptations of the taxonomy. After the workshop, selected materials—including the refined taxonomy, speculative designs, and their assessed qualities—will be shared on the workshop website and archived in an open-access repository (e.g., arXiv) (upon participant consent). These outputs also inform a post-workshop publication that revisits the quality of speculation in HCI. This publication will draw on workshop discussions, exemplary cases, and collected criteria to examine whether—and how—common ground on the quality of speculative design can be achieved within HCI.

3 Anticipated Outcomes

Collective Mapping: The workshop will produce a collective mapping of how quality in speculative design is understood across different contexts, illustrated with exemplary cases. This will highlight

both shared ground and productive disagreement. **Strengthening Tools and Practices:** Participants will provide feedback on the taxonomy, identifying where it supports the evaluation of speculative design and where it needs refinement. Building on this, the workshop will generate a starting point for refinements to the taxonomy and concrete recommendations for its use in research, teaching, and collaboration. Engaging with the taxonomy will strengthen participants’ ability to justify and assess speculative work by giving them a shared vocabulary, and practical experience in applying it to concrete speculations. This will help participants articulate evaluations more clearly in contexts such as peer review, teaching, and collaborative projects. **Community Building:** The workshop will foster an emerging network of researchers dedicated to advancing the understanding of quality in speculative design. Engaging with the taxonomy will provide participants with a shared reference point for discussing and evaluating speculative designs. To sustain exchange and collaboration beyond the workshop, we will set up a Discord group. **Publishing:** Workshop submissions and outcomes will be summarized and shared on the workshop website and archived in an open-access repository (e.g., ArXiv). We also plan a joint post-workshop publication, with possible dissemination through CHI, DIS or ACM Interactions.

4 Organizers

Ronda Ringfort-Felner is a PhD candidate at the University of Siegen. Her research combines speculative and performative approaches with design for well-being to explore how emerging technologies shape meaningful human–technology experiences. Her dissertation specifically investigates how speculative design can be practiced as a research method, with a particular focus on how its quality can be conceptualized and evaluated. She is co-editor of *Meaningful Futures with Robots* and has contributed to the field through publications such as *Design Fiction in a Corporate Setting: A Case Study*, and *The Quality of Speculation: A Scoping Review*, which introduced the descriptive taxonomy of quality criteria that forms the basis of this workshop.

Judith Dörrenbächer is a Postdoctoral Researcher at the University of Siegen whose work explores speculative and performative methods in design, techno-animism, and more-than-human perspectives. She investigates how speculative and critical design can question technological futures and open alternative imaginaries, with a particular focus on social robots. Judith is co-editor of *Meaningful Futures with Robot*, published widely on robots and speculative design, and co-organized workshops at leading HCI venues such as DIS.

Chris Elsdén is a Chancellor’s Fellow in Service Design in the Institute for Design Informatics at the University of Edinburgh and a pioneer of speculative enactments. He is a design researcher, with a background in sociology, and expertise in the human experience of data-driven services. Using and developing innovative design research methods, his work undertakes diverse, qualitative and often speculative engagements with participants to investigate emerging relationships with technology – particularly data-driven tools, FinTech and blockchain technologies.

James H. Auger is researcher-lecturer at the École normale supérieure (ENS Paris-Saclay). His work explores ways through which practice-based design research can lead to more considered

and democratic technological futures. James is co-founder of the speculative design practice Auger-Loizeau. Their projects have been published and exhibited internationally, notably at MoMA (NYC), Gallery 21_21 (Tokyo), at the Science Museum (London), the National Museum of China (Beijing) and at Ars Electronica (Linz). He has shaped the field through widely cited publications such as *Speculative Design: Crafting the Speculation*, *Beyond Speculative Design: Past–Present–Future*, and *Living with Robots: A Speculative Design Approach*.

James Pierce is an Associate Professor and Chair of Interaction Design in the Division of Design at the University of Washington (UW), and an Adjunct Associate Professor in the Human Centered Design and Engineering Department. His research examines how interfaces mediate human action, perception, and social practice, drawing on diverse theoretical perspectives. Methodologically, he combines techniques of design prototyping, qualitative methods, and critical perspectives to understand and address social issues connected to computing technology. James's work has advanced design as a mode of academic inquiry, participatory engagement, cultural critique, and speculative exploration.

Richmond Y. Wong is an Assistant Professor of Digital Media at Georgia Tech's School of Literature, Media, and Communication, where he directs the Creating Ethics Infrastructures Lab. His research focuses on understanding the social values, ethical issues, and work involved in technology production and use. Specifically, he studies how technology professionals address ethical issues in their work, and how to create the social and organizational conditions that can help support technologists to make ethical decisions. Richmond's work combines qualitative and design-based methods including speculative design and design fiction approaches.

Marc Hassenzahl is Professor of Ubiquitous Design at the University of Siegen and an internationally recognized scholar in experience design and human–technology interaction. His research bridges design theory, psychology, and speculative practices to explore how technologies can create meaningful and positive experiences. Marc has published extensively on design for well-being, and the role of speculative design in shaping meaningful futures. He regularly serves as organizer, keynote speaker, and program committee member at major HCI conferences.

5 Call for Participation

Speculative design has become a key approach in HCI for exploring the “not-yet.” Yet what counts as *good* speculation—and how to conceptualize its quality—remain open questions. This two-session workshop invites researchers, designers, and practitioners from diverse backgrounds to explore questions of quality in speculative design. Building on a recently published descriptive taxonomy of nine quality criteria, the workshop offers this framework as an *initial* common ground for reflection and critique. Participants will analyze speculative designs and critically discuss whether—and how—shared understandings of quality can be established.

We welcome submissions from anyone engaging with speculative practices or interested in what quality means in this context. Submissions should include a speculative design (e.g., fictional research paper, scenario description, or documentation of an enactment) along with a short reflection on what makes it strong or weak

as a speculation, in relation to its goals. The speculation may be the participant's own work or one they wish to discuss. Accepted formats include position papers (up to four pages, SIGCHI format), pictorials, posters, or other creative formats. Participants should avoid reading the taxonomy in advance to reflect from their own practice.

Submissions should be sent via: <http://www.qualityofspeculation.com>. Selection will be based on relevance and diversity of perspectives. Accepted contributions will be published on the workshop website and inform a post-workshop publication. If accepted, at least one author must register for and attend the workshop at CHI 2026. We expect 12–15 participants to enable focused discussion and intensive group work.

Acknowledgments

This work was partially funded by the German Federal Ministry of Research, Technology and Space (BMFTR) as part of the research program Interactive Technologies for Health and Quality of Life (Grant 16SV9108). Additional support was provided by the UK EPSRC Next Stage Digital Economy Centre in the Decentralised Digital Economy (DECaDE) (EP/T022485/1) for Chris Elsdén; by the National Science Foundation (#2142795) for James Pierce; and by the Atlanta Interdisciplinary AI Network for Richmond Y. Wong.

References

- [1] James Auger. 2013. Speculative design: Crafting the speculation. *Digital Creativity* 24, 1 (March 2013), 11–35. <https://doi.org/10.1080/14626268.2013.767276>
- [2] Jeffrey Bardzell and Shaowen Bardzell. 2013. What is “Critical” about Critical Design? In *SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*, 2013. Association for Computing Machinery, 3297–3306. <https://doi.org/10.1145/2470654.2466451>
- [3] Jeffrey Bardzell, Shaowen Bardzell, and Erik Stolterman. 2014. Reading critical designs: Supporting reasoned interpretations of critical design. In *Conference on Human Factors in Computing Systems - Proceedings, 2014. Association for Computing Machinery, 1951–1960*. <https://doi.org/10.1145/2556288.2557137>
- [4] Shaowen Bardzell, Jeffrey Bardzell, Jodi Forlizzi, John Zimmerman, and John Antanitis. 2012. Critical design and critical theory: The challenge of designing for provocation. In *Proceedings of the Designing Interactive Systems Conference, DIS '12*, 2012. 288–297. <https://doi.org/10.1145/2317956.2318001>
- [5] Eric P.S. Baumer, Mark Blythe, and Theresa Jean Tanenbaum. 2020. Evaluating design fiction: The right tool for the job. In *DIS 2020 - Proceedings of the 2020 ACM Designing Interactive Systems Conference, July 03, 2020. Association for Computing Machinery, Inc, 1901–1913*. <https://doi.org/10.1145/3357236.3395464>
- [6] Julian Bleeker. 2009. *Design Fiction: A short essay on design, science, fact and fiction*. Near Future Laboratory (2009).
- [7] Mark Blythe and Enrique Encinas. 2016. The Co-Ordinates of Design Fiction: Extrapolation, Irony, Ambiguity and Magic. In *Proceedings of the 2016 ACM International Conference on Supporting Group Work (GROUP '16)*, 2016. Association for Computing Machinery, New York, NY, USA, 345–354. <https://doi.org/10.1145/2957276.2957299>
- [8] Paul Coulton, Joseph Lindley, Miriam Sturdee, and Michael Stead. 2017. Design Fiction as World Building. *Proceedings of the Research Through Design Conference March (2017)*, 164–179. <https://doi.org/10.6084/m9.figshare.4746964>
- [9] Jan Derboven and Bert Vandenbergh. 2016. NewSchool: Studying the effects of design fiction through personalized learning scenarios. In *ACM International Conference Proceeding Series, October 23, 2016. Association for Computing Machinery*. <https://doi.org/10.1145/2971485.2993926>
- [10] Anthony Dunne and Fiona Raby. 2013. *Speculative Everything. Design, Fiction, and Social Dreaming*. MIT Press.
- [11] Chris Elsdén, David Chatting, Abigail C. Durrant, Andrew Garbett, Bettina Nissen, John Vines, and David S. Kirk. 2017. On speculative enactments. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*, 2017. Association for Computing Machinery, 5386–5399. <https://doi.org/10.1145/3025453.3025503>
- [12] Pedro Gil Farias, Roy Bendor, and Bregje F. Van Eekelen. 2022. Social dreaming together: A critical exploration of participatory speculative design. In *ACM International Conference Proceeding Series, August 19, 2022. Association for Computing Machinery*, 147–154. <https://doi.org/10.1145/3537797.3537826>

- [13] Gabriele Ferri, Jeffrey Bardzell, Shaowen Bardzell, and Stephanie Louraine. 2014. Analyzing critical designs: Categories, distinctions, and canons of exemplars. In *Proceedings of the Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques, DIS, 2014*. Association for Computing Machinery, 355–364. <https://doi.org/10.1145/2598510.2598588>
- [14] Christopher Frayling. 1993. *Research in Art and Design*.
- [15] Derek Hales. 2013. Design fictions an introduction and provisional taxonomy. *Digital Creativity* 24, 1 (March 2013), 1–10. <https://doi.org/10.1080/14626268.2013.769453>
- [16] Netta Iivari and Kari Kuutti. 2017. Critical Design Research and Information Technology: Searching for Empowering Design. In *Proceedings of the 2017 Conference on Designing Interactive Systems (DIS '17)*, 2017. Association for Computing Machinery, New York, NY, USA, 983–993. <https://doi.org/10.1145/3064663.3064747>
- [17] Eva Knutz, Thomas Markussen, and Poul Rind Christensen. 2014. The Role of Fiction in Experiments within Design, Art & Architecture - Towards a New Typology of Design Fiction. *Artifact* 3, 2 (December 2014), 8. <https://doi.org/10.14434/artifact.v3i2.4045>
- [18] Sandjar Kozubaev, Chris Elsdén, Noura Howell, Marie Louise Juul Søndergaard, Nick Merrill, Britta Schulte, and Richmond Y Wong. 2020. Expanding Modes of Reflection in Design Futuring. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020. 1–15. <https://doi.org/10.1145/3313831.3376526>
- [19] Joseph Lindley and Paul Coulton. 2015. Back to the Future: 10 Years of Design Fiction. In *Proceedings of the 2015 British HCI Conference*, 2015. Association for Computing Machinery, 210–211. <https://doi.org/10.1145/2783446.2783592>
- [20] Joseph Lindley, Paul Coulton, and Emmett L. Brown. 2016. Peer review and design fiction: “Great scott! the quotes are redacted.” In *Conference on Human Factors in Computing Systems - Proceedings*, May 07, 2016. Association for Computing Machinery, 583–594. <https://doi.org/10.1145/2851581.2892568>
- [21] Joseph Lindley, Dhruv Sharma, and Robert Potts. 2015. Operationalizing Design Fiction with Anticipatory Ethnography. *Ethnographic Praxis in Industry Conference Proceedings* 2015, 1 (October 2015), 58–71. <https://doi.org/10.1111/1559-8918.2015.01040>
- [22] Matt Malpass. 2013. Between wit and reason: Defining associative, speculative, and critical design in practice. *Design and Culture* 5, 3 (November 2013), 333–356. <https://doi.org/10.2752/175470813X13705953612200>
- [23] Thomas Markussen and Eva Knutz. 2013. The Poetics of Design Fiction. In *Proceedings of the 6th International Conference on Designing Pleasurable Products and Interfaces (DPPI '13)*, 2013. Association for Computing Machinery, New York, NY, USA, 231–240. <https://doi.org/10.1145/2513506.2513531>
- [24] Ivica Mitrović, Oleg Šuran, Matt Ward, Julian Hanna, James Auger, Enrique Encinas, Ingi Helgason, Michael Smyth, and Sara Božanić. 2009. *Beyond Speculative Design: Past – Present – Future*. SpeculativeEdu; Arts Academy, University of Split.
- [25] Larissa Vivian Nägele, Merja Ryöppy, and Danielle Wilde. 2018. PDFI: Participatory design fiction with vulnerable users. In *ACM International Conference Proceeding Series*, September 29, 2018. Association for Computing Machinery, 819–831. <https://doi.org/10.1145/3240167.3240272>
- [26] James Pierce. 2021. In tension with progression: Grasping the frictional tendencies of speculative, critical, and other alternative designs. In *Conference on Human Factors in Computing Systems - Proceedings*, May 06, 2021. Association for Computing Machinery. <https://doi.org/10.1145/3411764.3445406>
- [27] James Pierce, Phoebe Sengers, Tad Hirsch, Tom Jenkins, William Gaver, and Carl Disalvo. 2015. Expanding and refining design and criticality in HCI. In *Conference on Human Factors in Computing Systems - Proceedings*, April 18, 2015. Association for Computing Machinery, 2083–2092. <https://doi.org/10.1145/2702123.2702438>
- [28] Ronda Ringfort-Felner, Judith Dörrenbächer, and Marc Hassenzahl. 2025. The Quality of Speculation-A Scoping Review. In *Designing Interactive Systems Conference (DIS '25)*, July 2025. Funchal, Portugal. <https://doi.org/10.1145/3715336.3735794>
- [29] Ronda Ringfort-Felner, Robin Neuhaus, Judith Dörrenbächer, Sabrina Großkopp, Dimitra Theofanou-Fülbiér, and Marc Hassenzahl. 2023. Design Fiction in a Corporate Setting – a Case Study. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23)*, July 10, 2023. Association for Computing Machinery (ACM), 2093–2108. <https://doi.org/10.1145/3563657.3596126>
- [30] Britta F. Schulte, Paul Marshall, and Anna L. Cox. 2016. Homes for life: A design fiction probe. In *ACM International Conference Proceeding Series*, October 23, 2016. Association for Computing Machinery. <https://doi.org/10.1145/2971485.2993925>
- [31] Tim Seitz. 2017. *Design Thinking und der neue Geist des Kapitalismus*. transcript Verlag, Bielefeld, Germany. <https://doi.org/10.14361/9783839439630>
- [32] Bruce Sterling. 2012. Sci-Fi Writer Bruce Sterling Explains the Intriguing New Concept of Design Fiction (Interview by Torie Bosch). Slate. Retrieved from <https://slate.com/technology/2012/03/bruce-sterling-on-design-fictions.html>
- [33] Ron Wakkary, William Odom, Sabrina Hauser, Garnet Hertz, and Henry Lin. 2015. Material speculation: actual artifacts for critical inquiry. In *Proceedings of The Fifth Decennial Aarhus Conference on Critical Alternatives (CA '15)*, 2015. Aarhus University Press, Aarhus N, 97–108. <https://doi.org/10.7146/aaahcc.v1i1.21299>
- [34] Richmond Y. Wong. 2021. Using design fiction memos to analyze ux professionals' valueswork practices a case study bridging ethnographic and design futuring methods. In *Conference on Human Factors in Computing Systems - Proceedings*, May 06, 2021. Association for Computing Machinery. <https://doi.org/10.1145/3411764.3445709>
- [35] Richmond Y. Wong and Vera Khovanskaya. 2018. Speculative Design in HCI: From Corporate Imaginations to Critical Orientations. In *New Directions in Third Wave Human-Computer Interaction: Volume 2 - Methodologies*, Veronika Filimowicz Michael and Tzankova (ed.). Springer International Publishing, Cham, 175–202. https://doi.org/10.1007/978-3-319-73374-6_10